Image Classification with Feedforward Neural Networks Using Fashion-MNIST

1. Introduction

This report presents the design, training, and evaluation of a feedforward neural network (FFNN) for image classification using the Fashion-MNIST dataset. The assignment showcases how deep learning methods—particularly neural networks—can be applied to classify complex images more effectively than traditional machine learning models. The Fashion-MNIST dataset consists of 70,000 grayscale images of fashion items distributed across 10 categories, including shirts, coats, dresses, sneakers, and more. Each image has a resolution of 28×28 pixels. Because several classes share visual similarities (e.g., “T-shirt/top” vs. “Shirt”), the dataset provides a meaningful challenge for neural networks and serves as a benchmark for evaluating the strengths of deep learning models.

All implementation steps—including data preparation, preprocessing, neural network construction, training, and evaluation—were completed in Google Colab.

2. Methodology

2.1 Data Loading and Preprocessing

The Fashion-MNIST dataset was loaded using TensorFlow/Keras, automatically providing separate training and test sets of 60,000 and 10,000 images respectively. Sample images were visualized to understand class variety.

Preprocessing included:

- Normalization: Pixel values scaled to the [0,1] range.

- Reshaping: Adding a final dimension to convert image shapes into (28, 28, 1).

- One-hot Encoding: Transforming integer labels into categorical vectors.

- Train/Validation Split: 90% training, 10% validation.

2.2 Data Augmentation

To reduce overfitting and improve generalization, training images were augmented using:

- Random rotations (±15°)

- Width and height shifts (±10%)

- Horizontal flips

2.3 Neural Network Architecture

A simple feedforward neural network (FFNN) was constructed with:

- Flatten Layer
- Dense Layer (256 units, ReLU)
- Dropout (0.3)
- Dense Layer (128 units, ReLU)
- Dropout (0.3)
- Dense Output Layer (10 units, Softmax)

Compiled with Adam optimizer and categorical crossentropy. Total parameters: 235,146.

2.4 Training Process

The model was trained for 15 epochs with batch size 64. Validation accuracy improved steadily, indicating stable convergence.

3. Results and Analysis

3.1 Test Performance

- Test Accuracy: 0.8197
- Test Loss: 0.4656
- Macro Precision: 0.8174
- Macro Recall: 0.8197
- Macro F1-Score: 0.8106

3.2 Classification Insights

High-performing classes: Trouser, Bag, Sneaker, Ankle Boot.

Challenging classes: Shirt, Pullover, Coat due to visual similarities.

3.3 Confusion Matrix

The matrix revealed expected confusions among upper-body clothing.

3.4 Training and Validation Curves

Both accuracy and loss curves showed stable learning without overfitting.

4. Practical Applications and Deployment Considerations

Applications:

- Fashion retail automation
- Digital media organization
- Inventory classification

Challenges:

- Scalability and latency
- Model drift in evolving fashion trends
- Ambiguous class boundaries

5. Conclusion and Future Work

The FFNN achieved strong performance with 81.97% accuracy. Future improvements should include CNN architectures, hyperparameter optimization, transfer learning, and model compression for deployment.